SafeView

Intelligent Image Content Moderation Tool

COMP 264 — Cloud Machine Learning | Deliverable 4: Architecture Diagrams

Diagram 1 — Three-Tier Architecture Overview

Shows the full system stack: the browser client tier (frontend HTML/JS), the AWS Chalice serverless backend tier with its four Python modules, and the AWS managed services tier (S3, Rekognition, Comprehend). Arrows show the POST /analyze request flowing down and the JSON report returning to the results panel.

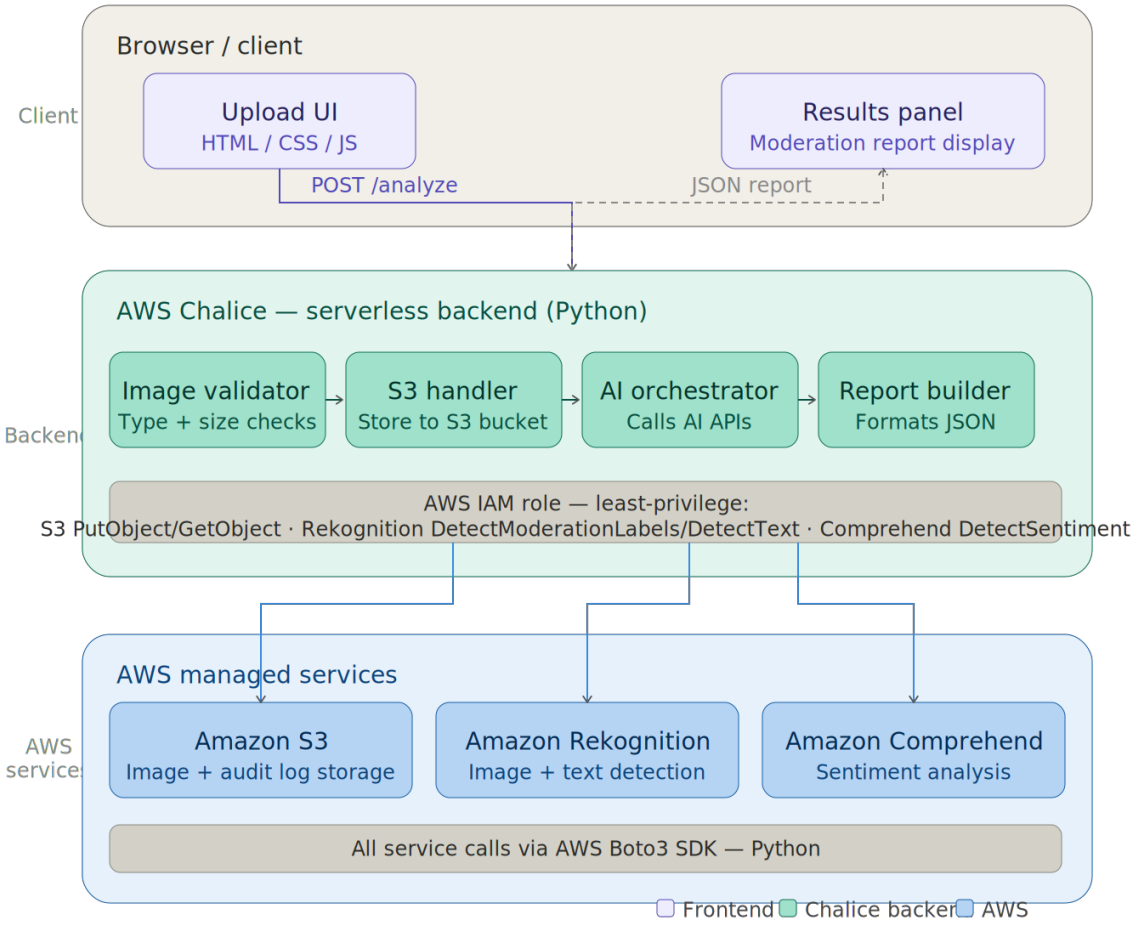

Figure 1: SafeView three-tier architecture — client, Chalice backend, and AWS AI services

Diagram 2 — REST Endpoints and AI Service Call Chain

Zooms into the two REST endpoints and the five-stage pipeline they trigger. POST /analyze runs the full pipeline (validate → S3 upload → Rekognition → Comprehend → report assembly). POST /analyze/threshold short-circuits to re-filter existing results without re-calling any AI service.

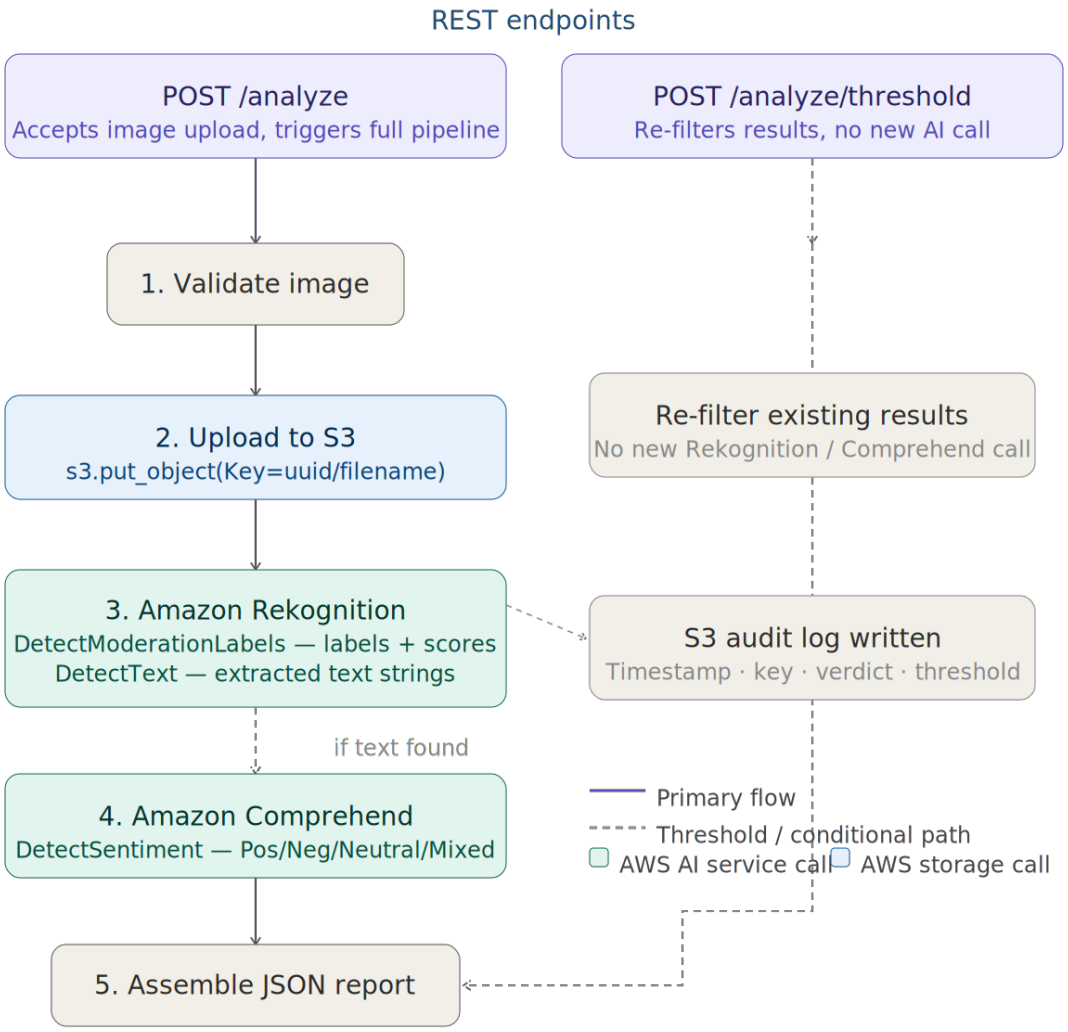

Figure 2: SafeView endpoint routing and AI stack call chain — solid lines = primary flow, dashed = conditional/threshold path

---

Architecture notes

- **AWS services:** Amazon Rekognition (DetectModerationLabels, DetectText) · Amazon Comprehend (DetectSentiment) · Amazon S3 · AWS IAM · AWS Lambda (via Chalice)
- **SDK:** All AWS calls made through the Boto3 Python SDK
- **IAM policy:** Least-privilege role — only the exact actions required by each module are granted
- **Multi-user support:** Each request uses a UUID-prefixed S3 key (uploads/{uuid}/filename) to prevent collisions
- **Design pattern:** Separation of Concerns — each module (validator, S3 handler, AI orchestrator, report builder) has a single responsibility